# Cornerstones or Stumbling Blocks? Deciphering the ‘Rock Tokens’ in On-Policy Distillation

# Step 1: Identify the Recalcitrant Tokens

Recalcitrant Tokens：Tokens that maintain **High Loss** before and after the entire On-Policy Distillation (OPD) process.

Off policy

To... the nearest hundredth of a minute.Ċ</think>ĊĊTo solve this problem, let's define the variables and set up the equations based on the given information.ĊĊ**Let:**Ċ- \( v \) = speed of the ship in still water (in km/h)Ċ- \( c \) = speed of the river (in km/h)ĊĊ**Given:**Ċ1. The ship travels 24 km upstream and 28 km downstream.Ċ2. It takes half an hour less than traveling 30 km upstream and 21 km downstream.Ċ3. It takes half an hour more than traveling 15 km upstream and 42 km downstream.ĊĊ**Step 1: Translate the information into equations.**ĊĊThe time taken to travel a certain distance is given by:Ċ\[ \text{Time} = \frac{\text{Distance}}{\text{Speed}} \]ĊĊ**Equation 1 (24 km upstream and 28 km downstream):**Ċ\[ \frac{24}{v - c} + \frac{28}{v + c} = T \]Ċwhere \( T \) is the total time.ĊĊ**Equation 2 (30 km upstream and 21 km downstream):**Ċ\[ \frac{

Hover over a token to see the original token string and its reverse KL value. Darker red means larger divergence.

# exp 1: OPD Experiment setup: (Zhao)

Students: Qwen3-4B-Instruct, llama 3.2 ,

Teachers: Qwen3-30B-Instruct,

# Experiment Setup

Teacher: qwen3-30b-instruct-2507

Student: qwen3-4b-instruct-2507

1. Stage 1: Off-line warm up. Using 20k teacher traces from openthougts3 to sft
2. Stage 2: RL (OPD or ours) 5k prompt, 4 rollouts per prompt

Checkpoints: https://huggingface.co/RockToken

# Result

| | Math 500 | aime25 |
|---|---|---|
| Qwen3-4b (base) | | |
| +sft(warmup) | | |
| +sft+opd | | |
| | | |

# exp 2: Rock Tokens（esp. pillar）Identification（Chao）

Dataset: MATH500, Sample: 100

| Rank | Token ID | Token String | Avg KL | Frequency |
|---|---|---|---|---|
| 1 | 9309 | 'when' | 7.471422 | 1 |
| 2 | 36340 | 'East' | 6.949764 | 1 |
| 3 | 3053 | 'translation' | 6.877148 | 1 |
| 4 | 6693 | ' whose' | 6.598140 | 2 |
| 5 | 45620 | ' rotated' | 6.551281 | 1 |
| 6 | 61165 | ' denomin' | 6.395685 | 1 |
| 7 | 2299 | "'re" | 5.718763 | 2 |
| 8 | 14257 | ' derived' | 5.548911 | 1 |
| 9 | 57134 | 'better' | 5.327905 | 1 |
| 10 | 33773 | ' frequencies' | 4.908598 | 2 |
| 11 | 23130 | ' representatives' | 4.681464 | 2 |
| 12 | 3953 | ' particular' | 4.647254 | 1 |
| 13 | 1831 | 'miss' | 4.312062 | 1 |
| 14 | 501 | ' new' | 4.295489 | 1 |
| 15 | 12140 | ' vertical' | 3.599760 | 1 |
| 16 | 15292 | 'vertical' | 3.412373 | 2 |
| 17 | 37591 | 'factor' | 3.389455 | 1 |
| 18 | 943 | ' type' | 3.298891 | 1 |
| 19 | 61554 | 'steel' | 3.293072 | 1 |
| 20 | 12942 | ' resulting' | 3.167676 | 1 |
| 21 | 9519 | ' mixed' | 3.160565 | 2 |
| 22 | 4803 | 'like' | 3.150318 | 1 |
| 23 | 4057 | ' remove' | 3.147902 | 1 |
| 24 | 14137 | ' AB' | 3.147507 | 3 |
| 25 | 5622 | ' bottom' | 2.987448 | 1 |
| 26 | 24660 | ' polar' | 2.974201 | 3 |
| 27 | 9889 | 'original' | 2.923723 | 1 |
| 28 | 1634 | ' As' | 2.841268 | 5 |
| 29 | 27283 | ' minus' | 2.826727 | 3 |
| 30 | 28316 | ' lengths' | 2.790590 | 3 |
| 31 | 76833 | ' perpendicular' | 2.772989 | 1 |
| 32 | 7661 | 'More' | 2.686198 | 1 |
| 33 | 4792 | 'That' | 2.645232 | 4 |
| 34 | 9931 | ' reverse' | 2.616442 | 1 |
| 35 | 49143 | ' selects' | 2.582609 | 2 |
| 36 | 19788 | ' (\\' | 2.574967 | 4 |
| 37 | 55597 | 'cccc' | 2.554373 | 1 |
| 38 | 4106 | 'Eng' | 2.551910 | 1 |
| 39 | 11831 | ' Properties' | 2.444303 | 1 |
| 40 | 2677 | ' always' | 2.392399 | 1 |
| 41 | 45411 | 'cube' | 2.391839 | 1 |
| 42 | 6099 | ' remember' | 2.374856 | 1 |
| 43 | 653 | ' do' | 2.358994 | 2 |
| 44 | 2284 | 'ways' | 2.333525 | 1 |
| 45 | 38293 | 'along' | 2.309934 | 2 |
| 46 | 7857 | ' numer' | 2.299637 | 1 |
| 47 | 12555 | 'what' | 2.267390 | 1 |
| 48 | 6524 | ' letter' | 2.263427 | 1 |
| 49 | 64527 | 'Bring' | 2.244485 | 2 |
| 50 | 14413 | ' Ca' | 2.239279 | 2 |

```
'}'   ' +'  ' \\' 'frac' '{'    '1'   '}{'   'b'    '}'    ' +'  ' \\' 'frac' '{'    '4'   '}{'  'c'    '}'   ' +'
0.000 0.000 0.000 0.000  -0.000 0.000 -0.000 -0.000 -0.000 0.000 0.000 0.000  -0.000 0.000 0.000 -0.000 0.000 0.000

' \\' 'frac' '{'    '1'    '6'    '}{'  'd'    '}\n' '$$'  '\n\n' '---\n\n' '###' ' Step' ' '   '1'   ':'   ' Use'
0.000 0.000  -0.000 -0.000 -0.000 0.000 -0.000 0.000 0.000 0.000  0.001     0.000 0.008   0.000 0.000 0.000 0.002

' the' ' method' ' of' ' Lag' 'range' ' multi' 'pliers' ' ('  'or'  ' Ca' 'uch' 'y'   '-S'  'ch'  'warz' ' inequality'
0.051  0.247     0.000 0.006  0.000   0.334    0.000    0.806 0.063 3.080 0.000 0.000 0.000 0.000 0.000  2.567

')\n\n' 'This' ' is' ' a'  ' constrained' ' optimization' ' problem' ' with' ' a'  ' sum' ' of' ' recip' 'roc' 'als'
0.462   0.116  0.256 0.004 0.441          0.016           0.001      0.281   0.518 1.152  2.632 0.070    0.000 0.000

' and' ' a'  ' linear' ' constraint' '.'   ' A'  ' powerful' ' tool' ' here' ' is' ' the' ' **' 'Ca'  'uch' 'y'   '-S'
3.995  1.695 0.372     0.000         0.040 0.031 0.205       0.351   0.221   0.000 0.029  0.000 0.067 0.000 0.000 0.000

'ch'  'warz' ' inequality' '**'  ' in' ' the' ' **' 'Eng' 'el'  ' form' '**'  ' ('  'also' ' known' ' as' ' Tit' 'u'
0.000 -0.000 0.368         0.833 0.018 0.117  0.051 2.552 0.001 0.070   0.007 0.000 0.002  0.002    0.000 0.537  0.000

"'s"  ' Lemma' '):\n\n' '$$'  '\n'  '\\'  'frac' '{x'  '^'   '2'    '}{'  'a'   '}'   ' +'  ' \\'  'frac' '{'   'y'
0.027 0.342    0.219    0.839 0.000 0.000 0.000  0.000 5.199 -0.000 0.000 0.001 0.001 0.000 -0.000 0.000  0.000 0.000
```

Observation: Low KL at equations, High KL at wording of reasoning and some punctuations (eg. ‘**’)

# Systematic Approach (Rock Token)

**2. The Beta-Weighted Geometric Mean (Customizable Balance)**

If you want granular control over how much frequency matters versus magnitude, you can normalize both metrics and combine them using an exponentiated geometric mean. This operates similarly to an F1 score in classification.

$$\text{Rock Score}_i = \left(\frac{f_i}{\max(f)}\right)^{\alpha} \times \left(\frac{\overline{KL}_i}{\max(\overline{KL})}\right)^{\beta}$$

- $\alpha$ controls the importance of frequency.
- $\beta$ controls the importance of the divergence spike.
- Setting $\alpha = 0.3$ and $\beta = 0.7$ ensures you prioritize high-loss tokens, but forcibly down-weights any token that doesn't appear frequently.

# Systematic Approach (Rock Token)

**3. Bayesian Averaging (Laplace Smoothing for Loss)**

This is the most statistically rigorous approach. It pulls the average KL of rare tokens down toward the global average KL, preventing low-frequency outliers from dominating the top 20 list.

$$\text{Bayesian KL}_i = \frac{(f_i \times \overline{KL_i}) + (C \times \mu)}{f_i + C}$$

- $f_i$ is the frequency of token $i$.
- $\overline{KL_i}$ is the actual average KL of token $i$.
- $C$ is a confidence constant (e.g., the median frequency of all tokens).
- $\mu$ is the global average KL across the entire vocabulary.

If $f_i$ is very small, the score collapses to $\mu$ (the global average). If $f_i$ is very large (a true rock token), the score approaches its true $\overline{KL_i}$.

# Systematic Approach (Pillar/Stumbling Block)

**Entropy Filtering:** Calculate the Shannon entropy of the teacher's probability distribution for each rock token.

- *High Teacher Entropy* + *High KL Divergence:* The teacher is uncertain or allowing multiple paths, but the student is picking one. Likely a **Stumbling Block** (stylistic or branching).
- *Low Teacher Entropy* + *High KL Divergence:* The teacher is absolutely certain this specific token must come next, but the student disagrees. Likely a **Pillar**.

**Attention-Weight Probing:** Extract the self-attention weights from the student model during generation.

- If a rock token has high outgoing attention weights connecting it to the original mathematical constraints in the prompt, it is functionally interacting with the logic. It is a Pillar.
- If it only attends to local stylistic context, it is a Stumbling Block.

# 4.16-22 plan

## Step 2: Pillars or Stumbling Blocks：Causal Impact of Recalcitrant Tokens

For these Recalcitrant Tokens, we prohibited models from outputting them, and observe both the output format and reasoning performance changes.

Pillar: if removed, model training collapse

Stumbling rocks: if removed, models learn faster and better

**Problem Formulation:**

**Input:**

**Output:**

**Naive approach:**

Rock token: Manually cherry-pick some highly frequent or visually salient tokens as rock tokens.

Pillar tokens:  further cherry-pick tokens that appear semantically irrelevant to reasoning (e.g., “note”, “information”), especially those occurring at sentence beginnings or as standalone transition phrases.

1. Replace or mask selected pillar tokens during generation (e.g., substitute with meaningless tokens or block them from the vocabulary).
2. Measure the impact on downstream inference performance.
3. Compare against a baseline where an equal number of random tokens are masked.

**One step optimization:**

# Top 20 rock tokens

| Tokens | frequency | avg.loss |
| --- | --- | --- |

exp 3: Detect and remove stumbling rocks, and see whether pillar will rebuild? (Dipta)

1. Cross model token removal [1]
2. Stumbling tokens detection

3. Observe new pillar tokens (if exist)

[1] Multi-Level Optimal Transport for Universal Cross-Tokenizer Knowledge Distillation on Language Models

# Problem formulation-Stumbling Tokens Detection

**Input:**

**Output:**

**Naive approach:**

1. Manually cherry-pick a set of candidate tokens from the vocabulary.
2. set selected tokens' logits to `-inf`, during training. [2]
3. Run OPD (on-policy distillation) and observe whether the student model learns faster (e.g., improved convergence speed or performance).

**One step optimization:**

[2] Do LLMs Encode Functional Importance of Reasoning Tokens ?

# Problem formulation-Observe new pillar tokens

**Input:**

**Output:**

**Naive approach:**

1. Compute token-level statistics before and after training, especially **frequency** and **average loss**. Rank tokens according to a simple rock-token criterion based on these statistics.
2. Compare the ranked lists before and after training, and identify tokens that newly appear or rise substantially in the post-training rock-token set. Treat these as candidate **new rock tokens**.
3. Then test whether they are also **pillar tokens** by masking or replacing them during generation and measuring whether reasoning performance drops.

**One step optimization:**

# Exp 4. Reasoning tokens vs. Stylish Token

Ideally, High loss token in OPD represents factual incorrect information.

However, sometimes they are just stylish token.

Detect and skip such effort on stylish token learning will benefit.

# 20 days left

| M | T | W | R | F | Sat | Sun |
|---|---|---|---|---|---|---|
| 13th | | | Today<br>Zhao: evaling 3 ckpt on amie. | | | |
| 20th | | | | | | |
| 27th | | | | | | |
| 4th<br>Abstract deadline AOE | | 6th<br>Full paper deadline AOE | | | | |